

Final Project Summary & Executive Deliverable (TB-L15)

Task ID: TB-L15 Date: 05-Sep Scope: End-to-End Loan Classification System

PROJECT COMPLETED

TOTAL TASKS EXECUTED

15 / 15

KNN TEST ACCURACY

81.50%

DECISION TREE ACCURACY

80.00%

INTEGRATION STATUS

Dual-Model GUI

1. TASK EXECUTION & TEAM MILESTONES

Task ID & Date	Core Objective & Responsibility	Assigned Lead	Deliverable Output	Status
TB-L01 - TB-L03 (03-Sep)	Requirements elicitation, task breakdown & team allocation	Keerti C, Sanjana M, Abdussamad S	Requirements Doc, Work Breakdown	Completed
TB-L04 - TB-L07 (03/04-Sep)	Data quality checks, cleaning, EDA, feature alignment	Aditya K, Sanjana M, Keerti C	Cleaned Dataset, EDA Charts	Completed
TB-L08 (04-Sep)	Blocker identification (EDA skew & model mathematics)	Full Project Team	Blocker Report & Risk Matrix	Resolved
TB-L09 (05-Sep)	KNN model development & tuning (k=8, Euclidean, Scaler)	Namrata D	Model Progress Report, PKL files	Completed
TB-L10 (05-Sep)	Decision Tree model development (max_depth=3, CART Gini)	Abdussamad S	Model Progress Report, PKL files	Completed
TB-L11 (05-Sep)	Model evaluation, diagnostics & head-to-head comparison	Abdussamad S	Performance Matrix & CM Diagnostics	Completed
TB-L12 - TB-L15 (05-Sep)	Checklist audit, Tkinter GUI, slides, presentation & demo	Team / Abdussamad S Lead	GUI_App.py, Final Summary PDF	Completed

2. MACHINE LEARNING BENCHMARK & STRATEGIC TRADE-OFFS

KNN Classifier (k = 8) • Low-Risk Engine

- **Accuracy:** 81.50% | **Precision (Approved):** 78.95%
- **False Approvals (FP):** 4 applicants (Minimal capital risk)
- **False Rejections (FN):** 33 applicants (Conservative recall: 31.25%)
- **Best Application:** Strict risk aversion, default mitigation.

Decision Tree (Depth = 3) • High-Capture Engine

- **Accuracy:** 80.00% | **Recall (Approved):** 60.42%
- **Captures:** 29 approvals (Near 2x volume over KNN)
- **Explainability:** Rooted on Credit_Score (30.42%), Loan_Amount (17.65%)
- **Best Application:** Portfolio expansion and compliance reporting.

3. BLOCKER RESOLUTION & TECHNICAL MITIGATIONS

- **Blocker 1 (EDA Interpretation & Target Imbalance):** Resolved by identifying class distribution skew early and establishing stratified 80/20 train/test splits, ensuring consistent model evaluation.
- **Blocker 2 (Mathematical Concepts in Splits & Distances):** Demystified **Gini Impurity** ($1 - \sum p_i^2$) to justify tree branch logic, and applied **StandardScaler** to eliminate magnitude bias in **Euclidean Distance** calculations ($\sqrt{\sum (p_i - q_i)^2}$) across financial metrics.

4. SYSTEM DEPLOYMENT & DELIVERED ARTIFACTS

Delivered Pipeline Artifacts: knn_model.pkl decision_tree_model.pkl scaler.pkl model_columns.pkl GUI_App.py

Operational Conclusion: Both predictive engines have been successfully embedded into the interactive Tkinter application (GUI_App.py). Loan underwriting officers can select either the *Conservative / Low-Risk Mode (KNN)* or the *Growth / Explainable Mode (Decision Tree)* depending on risk criteria and organizational targets.